رواد مصر الرقمية

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AssetX

B2B Asset Marketplace

Project Team:

1. Mohamed Ashraf
2. Mohamed Mostafa
3. Ezz Mohamed
4. Mohamed Yasser
5. Youssef Moataz

PROJECT DOCUMENTATION
1.0 MVP

Software Proposal

1.1 Introduction

AssetX is a web-based B2B marketplace that connects companies holding surplus, used, or obsolete business assets with companies looking to acquire them at lower-than-new cost. The platform combines three transaction models in one place — Fixed Price sales, competitive Auctions, and demand-driven RFQs — starting with the Electronics category. This document presents the project proposal, analysis, core requirements, and design direction, and serves as the analysis and engineering reference for the team's project in the Digital Egypt Pioneers Initiative.

1.2 Problem Statement

In Egypt, businesses with surplus or no-longer-needed assets struggle to sell them in an organized and efficient way. Selling depends on personal networks, brokers, traders, or disconnected marketplaces, making it difficult to reach the right buyers and recover true asset value. At the same time, companies seeking affordable business equipment struggle to find suitable products, compare options, and identify reliable corporate sellers. The result is a fragmented B2B asset market with a weak connection between corporate supply and demand.

1.3 Objectives

ID	Objective
O-01	Build one centralized, organized B2B asset marketplace accessible to Egyptian companies.
O-02	Support three transaction models in a single platform: Fixed Price, Auctions, and RFQs.
O-03	Establish B2B trust through company profiles and Admin-verified company badges.
O-04	Launch an MVP focused on the Electronics category with Seller, Buyer, and Admin sides.
O-05	Validate the business model with a pilot: 10 verified companies, published listings, created RFQs, and completed pilot transactions.
O-06	Validate the commission-based revenue model before scaling to more categories.

1.4 System Overview

AssetX has three sides. Sellers register their company, list single assets or batches, and sell via Fixed Price or Auctions while managing everything from a dashboard. Buyers search and filter listings, purchase at fixed prices, bid in auctions, or publish RFQs describing what they need and select among seller offers. Admins verify companies, moderate listings, monitor auctions and RFQs, and resolve disputes. The flow is: List Assets → Connect → Transact.

Software Analysis

2.1 Introduction

2.1.1 Problem Analysis and Motivation

Supply exists (upgrades, restructuring, and replaced equipment pile up in stores) but has no organized outlet; demand exists (startups and SMEs need affordable equipment) but has no trusted discovery channel. Brokers fill the gap with high friction and low transparency. AssetX is motivated by connecting these two sides directly: sellers recover value, buyers save cost, and both transact with verified corporate identities instead of anonymous classifieds.

2.1.2 Scope of the Project

In scope (MVP)	Out of scope
• Company registration with manual Admin verification • Asset/batch listings (Electronics) • Fixed-price purchase • Basic timed auctions with simulated refundable deposits • RFQs with offers • Search, filters, and sorting • Seller dashboard • Admin moderation and monitoring • Basic dispute handling • In-app and email notifications • Sandbox/simulated payments	• Native mobile app • AI pricing • Logistics • Advanced analytics • Advanced auction features • Live payment processing and real-money escrow • Inspection services • Automated verification • Multi-user company accounts • Arabic interface

2.1.3 Target User Groups

User class	Company types	Decision makers	Goal
Sellers (Supply)	Tech & software companies, call centers, growing enterprises, companies upgrading hardware	IT Managers / CTOs, Procurement, Operations, Founders	Sell surplus and recover value
Buyers (Demand)	Startups, SMEs, co-working spaces, training academies	Operations, Procurement Managers	Reliable equipment below new-price
Platform Admin	AssetX operations	—	Verify, moderate, monitor, resolve disputes

2.1.4 Definitions, Acronyms, and Abbreviations

Term	Meaning
RFQ	Request for Quotation
MVP	Minimum Viable Product
Batch	Assets sold as one unit
Verified company	Admin-approved
Deposit	Refundable auction guarantee, simulated in MVP
Sandbox payment	Simulated, no real money
B2B	Business-to-business
RBAC	Role-based access control

2.2 Related Work

2.2.1 Traditional Channels: Brokers and Personal Networks

The default channel today — relationship-driven, geographically limited, opaque pricing, no comparison, no records. It works for one-off deals and fails at scale.

2.2.2 Generic Marketplaces and Groups

OLX / dubizzle Egypt and OpenSooq Egypt (plus informal WhatsApp/Facebook groups) list equipment but target consumers, mix business assets with everything else, offer no auction or RFQ mechanics, and provide no corporate verification.

2.2.3 Existing Specialized Applications

Platform	Market	Focus
ACE Group	Egypt	Industrial assets + valuation
eMazad	Egypt	Online auctions + corporate asset sales
Ritchie Bros.	Global	Heavy equipment + auctions
IronPlanet	Global	Used equipment + inspection reports
MachineryTrader	Global	Marketplace + want-to-buy
Machinio	Global	Equipment search + buyer–seller connection

2.2.4 Limitations of Existing Solutions

Most competitors cover one side or one model: auctions only (eMazad, Ritchie Bros.), equipment listings only (IronPlanet, MachineryTrader, Machinio), or general listings (OLX, OpenSooq). None combines Fixed Price + Auctions + RFQs for verified corporate users in the Egyptian B2B context. AssetX's gap: one platform where companies both sell what they have and request what they need.

2.3 System Requirements

2.3.1 Project Sponsor

Developed within the Digital Egypt Pioneers Initiative; no external sponsor.
Initial pilot costs are near-zero (free-tier hosting, open-source stack).

2.3.2 Business Need

Egyptian companies need an organized channel to liquidate surplus assets and source affordable equipment with corporate-grade trust — reducing broker dependence and value loss on both sides.

2.3.3 Business Request

Build and pilot an MVP web platform where verified companies can list assets/batches, transact via Fixed Price, Auctions, and RFQs, and complete supervised pilot transactions — then validate the commission model before scaling.

2.3.4 Business Value

Sellers reach corporate buyers and recover asset value; buyers cut equipment costs and search time; AssetX earns transaction commission with future featured-listing and subscription streams. Strategic value: first-mover data on B2B asset supply/demand in Egypt.

2.3.5 Measuring Success

- 10 verified companies onboarded with 0% introductory commission
- Published listings across Electronics subcategories; RFQs created with seller responses
- Completed pilot transactions (fixed, auction, and RFQ paths each demonstrated)
- Performance: search/filter under 2 seconds; bid acknowledgement under 1 second
- Benefits: lower customer-acquisition cost per deal vs. brokers; reusable verified-company network; validated willingness to transact online

2.3.6 Specific Problem Addressed by This Build

The MVP specifically proves that strangers — verified companies with no prior relationship — can discover, negotiate (via price, bid, or RFQ offer), and complete an asset deal through the platform with simulated money movement.

2.4 Feasibility Study

Dimension	Assessment	Verdict
Technical	Required skills (web frontend, relational DB design, API development) substantially overlap the team's training; identified gaps are authentication/payment-sandbox integration and auction concurrency, mitigated by documented learning tasks and the sandbox-only constraint.	Feasible
Economic	Near-zero cash cost (open-source stack, free-tier hosting); main cost is team time. Revenue validation is the pilot's goal, not a precondition.	Feasible
Organizational	Five-member team with weekly reviews and a shared board; company onboarding depends on members' professional networks (noted as the critical path).	Feasible with active outreach
Marketing	Direct B2B outreach (LinkedIn, email, phone, WhatsApp) plus IT-provider and startup-community partnerships per the acquisition plan.	Feasible for a 10-company pilot

2.5 Interviews: Plan (Results Pending)

Customer interviews are scheduled as the first validation activity. Guide questions:

1. How do you currently dispose of surplus equipment?
2. What does that process cost you in time and fees?
3. How do you source used equipment today, and what goes wrong?
4. What would make you trust an online counterparty?
5. Which matters more: fixed price, auction, or requesting quotes?
6. What would stop you from completing a deal online?

Findings will be appended here before pilot launch.

2.6 Business Model

➢ Revenue Model

The primary revenue stream is a commission on every successful transaction completed through the platform.
The commission rate, and whether it is paid by the Seller, the Buyer, or split between both, will be determined through customer validation rather than assumed.
Future revenue streams include:
- Featured listings and promoted assets
- Premium company subscriptions
The initial goal is to validate the transaction flow and the commission model before scaling.

➢ First Users Strategy

To attract the first 10 verified companies, AssetX offers:
- 0% transaction commission during the pilot
- Free listings during the initial stage
- Direct access to relevant RFQs
- Simple and fast bulk listing
- Verified corporate users, to increase trust between counterparties

➢ User Acquisition Strategy

The main strategy is direct B2B outreach to decision-makers: IT Managers, CTOs, Procurement Managers, Operations Managers, and Startup Founders.
- Channels: LinkedIn, email, phone calls, and WhatsApp
- Partnerships: IT service providers, hardware distributors, technology companies, and startup communities
The acquisition path is: Direct Outreach → First Users → Initial Transactions → Referrals → Digital Marketing.

➢ Solving the Two-Sided Marketplace Problem

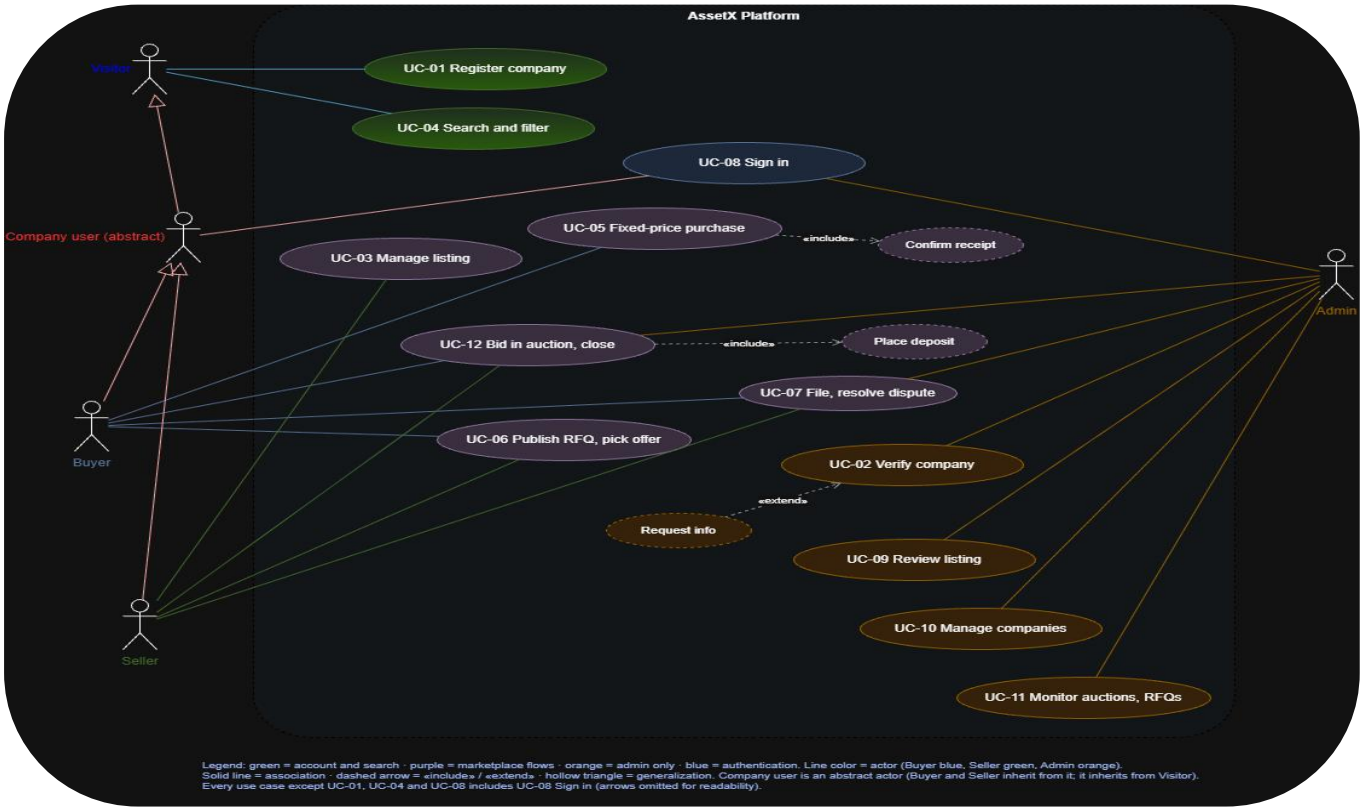
A marketplace needs sellers to attract buyers and buyers to attract sellers.
AssetX does not target the whole market at launch. I
t starts with one category (Electronics) and acquires the first sellers and buyers together through direct outreach.
Once real supply, demand, and transactions are proven, the platform expands to additional categories.

2.7 Use Cases

➢ Use Case Diagram

– The following diagram illustrates the main interactions between visitors, company users, buyers, sellers, and the platform administrator within AssetX.

Figure 2.1: AssetX Use Case Diagram



Actors: Seller Company (S), Buyer Company (B), Platform Admin (A), Visitor (V).

UC-01 — Register company (V)

Precondition	None
Flow	1. Visitor submits work email, password, company details 2. System validates and creates Unverified account 3. System sends verification email 4. User confirms email
Exception	Duplicate email or invalid data → system displays an error and does not create the account

UC-02 — Verify company (A)

Precondition	Pending company exists
Flow	1. Admin opens verification queue 2. Admin reviews details 3. Admin approves → public Verified badge (or rejects with reason) 4.Company is notified of the decision
Alternate	Request more information → status stays unverified

UC-03 — Manage listing (S)

Precondition	Verified company
Flow	1.Seller enters listing details, including title, description, category, condition, quantity, location, images, and sale type 2. Seller selects Fixed Price or Auction and enters the relevant pricing or auction details 3. Seller saves the listing as Draft or submits it for review 4.System validates the listing data 5.System sets the submitted listing to Pending Review until an Admin decides
Alternate	Seller edits a Draft or withdraws a listing before it is sold.
Exception	Missing required data → highlighted errors, nothing saved publicly

UC-04 — Search and filter assets (V/B)

Precondition	Published listings exist
Flow	1. Visitor or Buyer enters keyword and/or filters 2. System searches published listings and applies the selected filters 3. System displays matching results and allows sorting by the supported criteria 4. Visitor or Buyer opens a listing to view its details and seller company profile
Exception	No matching results → message suggesting to change the filters.

UC-05 — Fixed-price purchase (B)

Precondition	Verified Buyer, available published Fixed Price listing, and the listing does not belong to the Buyer company
Flow	1. Buyer confirms purchase 2. System reserves the listing instantly 3. System creates the Transaction (Pending) and holds the payment (simulated) → Payment Held 4. Both parties notified 5. Seller hands over the asset (outside the platform). 6. Buyer confirms receipt 7. System completes the Transaction and processes the simulated payment according to the approved commission policy 8. Listing becomes Sold
Alternate	Buyer cancels before payment is held → Transaction Cancelled, listing returns to Published
Exception	Listing already reserved or sold → message Simulated payment fails → Transaction Cancelled, listing returns to Published

UC-06 — Publish RFQ and select offer (B/S)

Precondition	Verified Buyer company
Flow	1. Buyer publishes needs with deadline 2. Sellers browse and submit offers 3. Buyer compares side by side 4. Buyer selects winner → transaction created 5. Losers notified
Alternate	Buyer cancels the RFQ before selection → RFQ Cancelled Deadline passes with no selection (or no offers) → RFQ Expired automatically
Exception	Offer submitted after the deadline → rejected Seller submits an offer on its own company's RFQ → rejected

UC-07 — File and resolve dispute (B/S/A)

Precondition	Existing Transaction involving the Buyer or Seller company
Flow	1. Buyer or Seller files a dispute with a reason → Dispute Open, Transaction Disputed 2. Admin reviews the dispute and the transaction details → Under Review 3. Admin records the resolution 4. Transaction becomes Completed or Cancelled and the Dispute Resolved 5. Both parties are notified

UC-08 — Sign in (S/B/A)

Precondition	Registered account with a confirmed email
Flow	1.User enters email and password. 2.System authenticates the user. 3.System opens the home page for the user's role (company or Admin).
Alternate	Unverified company → signed in with browse-only access
Exception	Wrong email or password → error message Email not confirmed → system asks the user to confirm it first Suspended account → access denied with a message

UC-09 — Review listing (A)

Precondition	A listing is in Pending Review
Flow	1. Admin opens the listing review queue 2. Admin checks the listing details and photos 3. Admin approves → listing becomes Published and appears in the marketplace (or rejects with a reason) 4. Seller is notified of the decision.
Alternate	Seller edits a rejected listing and resubmits it → back to Pending Review
Exception	Seller withdraws the listing before the decision → it leaves the queue (Withdrawn).

UC-10 — Manage company accounts (A)

Precondition	Admin signed in; the company exists
Flow	1. Admin searches for a company and opens its profile 2. Admin suspends the account with a reason, or reactivates a suspended one. 3. System blocks sign-in and hides the company's published listings while suspended; reactivation restores them 4. Company is notified.
Alternate	Admin only views the company details (verification status, listings, transactions) without changing anything.
Exception	Company has active auctions or open transactions → system lists them and asks the Admin to confirm before suspending.

UC-11 — Monitor auctions and RFQs (A)

Precondition	Admin signed in
Flow	1. Admin opens the monitoring view of active auctions and open RFQs 2. Admin checks bids, offers, and deadlines 3. When a Seller requests to cancel a started auction (BR-07), Admin approves or rejects the request 4.Affected parties are notified.
Alternate	Admin rejects the cancellation → the auction continues unchanged. Admin only observes → no action.
Exception	Auction already ended → cancellation not possible.

UC-12 — Bid in auction and close it (B/S/A)

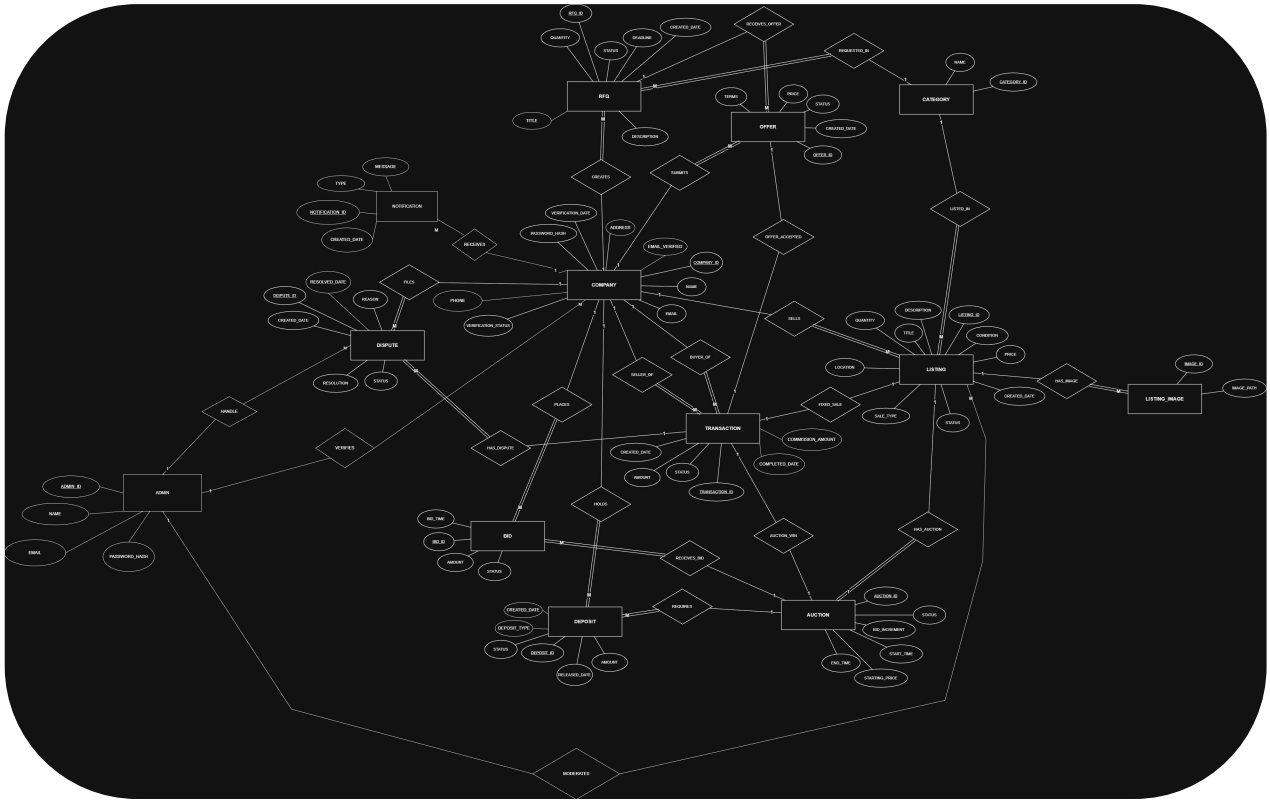
Precondition	Active auction; verified Buyer whose company does not own the listing
Flow	1. Buyer places the required deposit (held, simulated) 2. Buyer bids at least the starting price (first bid) or the current highest bid plus the increment 3. System validates the bid and updates the live auction board 4. Outbid Buyers are notified 5. At end time the system closes the auction automatically and marks the highest valid bid as Won 6. System creates the winner's Transaction (Pending) and continues as in UC-05 from payment held to receipt confirmation 7. Deposits are released after clean completion
Alternate	Seller cancels before the auction starts → Cancelled. After it starts, cancellation needs Admin approval
Exception	Bid late or below the minimum → rejected No held deposit → bidding not allowed No valid bids at end time → auction Ended and listing Expired

2.8 Database Design

2.8.1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) illustrates the main entities in AssetX and the relationships between them, providing the conceptual structure for the database design.

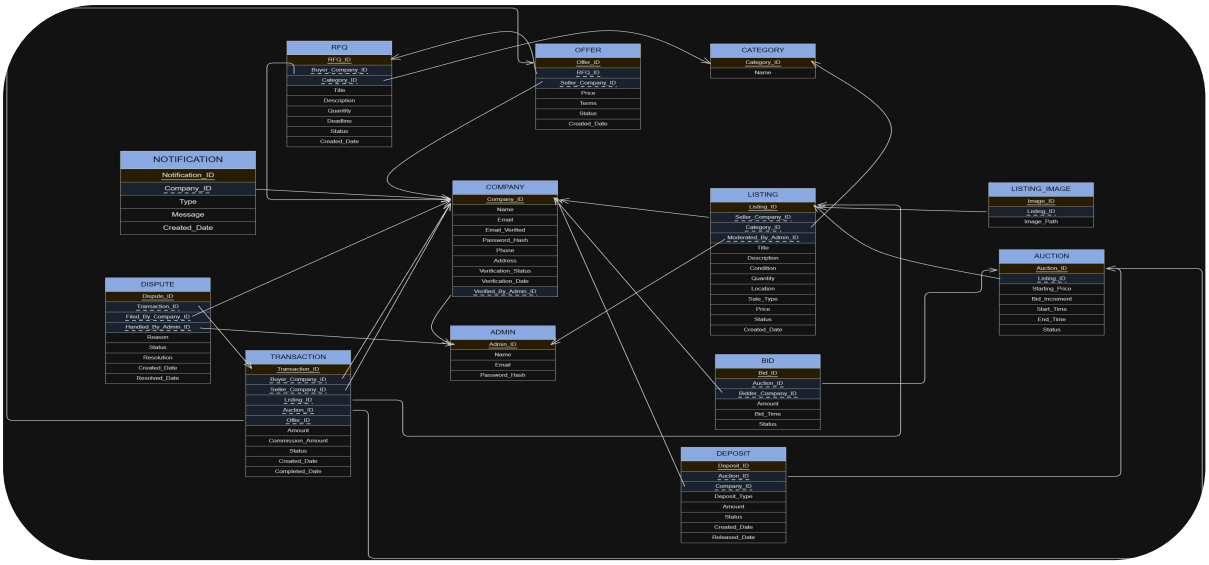
Figure 2.2: AssetX Entity Relationship Diagram (ERD)



2.8.2 Relational Mapping

The relational mapping translates the ERD into relational tables, identifying the attributes, primary keys (PKs), and foreign keys (FKs) used to represent the database structure.

Figure 2.3: AssetX Relational Mapping



2.9 Risks

ID	Risk	Likelihood	Impact	Mitigation
R-01	Not enough sellers and buyers at launch	High	High	Start with Electronics only. direct B2B outreach. 0% commission and free listings for the first 10 verified companies.
R-02	Companies reject the commission model	Medium	High	Validate the rate and who pays it during the pilot before fixing it
R-03	Low trust between companies(fake accounts,non delivery)	Medium	High	Admin verification before transacting. Payment held until the Buyer confirms receipt. Admin dispute resolution.
R-04	Simultaneous bids produce a wrong winner	Medium	High	Process bids consistently in the database. Test concurrent bidding. keep auction features basic in the MVP.
R-05	A winning bidder does not complete the purchase	Medium	Medium	Refundable deposit Define the deposit treatment during the pilot
R-06	MVP scope (Fixed Price, Auction, RFQ) is too large for the team	Medium	High	Build in order Fixed Price,then RFQ,then Auction. Defer advanced auction features.
R-07	Payments are simulated, so the real payment flow is untested	High	Medium	State it as an MVP limitation. Keep payment behind one service so a real gateway can replace the simulation later.

Software Requirement Specification

3.1 Requirement List

Type	Requirement Description
FR-1	The system shall allow a company to register an account with company details and login credentials.
FR-2	The system shall verify the company's email address; new companies remain Unverified until approved by an Admin.
FR-3	The system shall authenticate companies by email and password.
FR-4	The Admin shall review pending companies and approve or reject their verification, with a reason for rejection.
FR-5	A Seller shall create a listing for an asset or batch with title, description, category, condition, quantity, location, images, and sale type (Fixed Price or Auction).
FR-6	A Seller shall save a listing as a draft, publish it, edit it, and withdraw it; each listing has a status.
FR-7	The Admin shall review listings and approve or reject them before they become public.
FR-8	A Seller shall view a dashboard of their listings, auctions, received offers, and transactions.
FR-9	Visitors and buyers shall search published listings by keyword.
FR-10	Visitors and buyers shall filter and sort results by category, condition, location, and price.
FR-11	Visitors and buyers shall view listing details and the seller company profile.
FR-12	A verified Buyer shall purchase an available fixed-price listing.
FR-13	On purchase, the system shall reserve the listing and create a Transaction with the payment held (simulated).
FR-14	When the Buyer confirms receipt, the system shall complete the Transaction and process the simulated payment according to the approved commission policy
FR-15	The system shall track each Transaction status from creation to completion or cancellation and show it to both parties.
FR-16	A Seller shall create a timed auction for a listing with a starting price, bid increment, and start and end time.
FR-17	A verified Buyer shall bid only in an active auction and only after a deposit is held.
FR-18	The system shall record each valid bid (bidder, amount, time) and reject bids below the minimum or after the auction ends.
FR-19	The system shall close the auction automatically at its end time, identify the winning bid, and create the Transaction.
FR-20	The system shall release or refund deposits according to the auction outcome.
FR-21	A verified Buyer shall create an RFQ with description, category, quantity, and deadline.
FR-22	Sellers shall view open RFQs and submit offers (price and terms); the system records each offer with its Seller.
FR-23	The Buyer shall compare the received offers and select one; the selection creates a Transaction and marks the other offers as not selected.
FR-24	The Buyer may cancel an RFQ before selecting an offer; the system closes RFQs automatically at their deadline.

FR-25	A Buyer or Seller shall file a dispute with a reason on an existing Transaction.
FR-26	The Admin shall review disputes, record the resolution, and update the Transaction status accordingly.
FR-27	The Admin shall manage company accounts, including viewing them and suspending or reactivating them.
FR-28	The Admin shall monitor auctions and RFQs.
FR-29	The system shall notify users (in-app and by email) of verification, transaction, bidding, RFQ, and dispute events.
NFR-30	Search and filter results shall be returned in under 2 seconds at pilot scale.
NFR-31	A submitted bid shall be acknowledged in under 1 second.
NFR-32	Passwords shall be stored hashed, never in plain text, and all traffic shall use HTTPS.
NFR-33	Access shall be role-based (RBAC): only verified companies can transact, and only Admins can access administration functions.
NFR-34	Simultaneous bids shall be processed consistently so the highest valid bid wins, and a listing shall never be sold more than once.
NFR-35	The system shall be usable on desktop browsers (Chrome, Edge, Firefox) at the MVP stage.

3.2 Business Rules

1. AssetX is B2B-only: only registered companies can hold accounts and transact; individuals cannot register.
2. A company must be verified (Admin-approved) before it can sell, buy, bid, or create RFQs and offers; unverified companies can only browse.
3. In the MVP, listings and RFQs are limited to the Electronics category.
4. A company may act as both Buyer and Seller, but may not purchase, bid on, or submit offers against its own listings and RFQs.
5. An auction sells the complete asset or batch; partial sales are not allowed.
6. A bid is valid only if it meets the starting price (first bid) or exceeds the current highest bid by at least the bid increment.
7. A Buyer must hold a refundable deposit to bid in an auction; a Seller deposit prevents cancelling the auction once it starts (afterwards cancellation requires Admin approval).
8. If a winning bidder does not complete the purchase, the treatment of their deposit is to be defined during pilot validation.
9. An RFQ may receive multiple offers; only one offer is selected and the rest are marked as not selected.
10. A commission applies to every successful transaction; the rate and who pays it (Seller, Buyer, or split) are to be determined through customer validation.
11. During the pilot, the first 10 verified companies pay 0% commission and list for free.
12. Payments and deposits are simulated (sandbox) in the MVP; no real money is processed.

3.3 Status Lifecycle

- Company:

  Unverified → Verified (or Rejected, Suspended)

- Listing:

  Draft → Pending Review → Published → Reserved → Sold (or Rejected, Withdrawn, Expired)

- Auction:

  Scheduled → Active → Ended (or Cancelled)

- Bid:

  Active → Outbid or Won

- Deposit:

  Held → Released (or Forfeited)

- RFQ:

  Open → Closed (or Cancelled, Expired)

- Offer:

  Submitted → Selected or Not Selected

- Transaction:

  Pending → Payment Held → Completed (or Cancelled, Disputed)

- Dispute:

  Open → Under Review → Resolved

3.4 Acceptance Criteria

ID	Criteria
AC-01	A company can register with valid data and receives a verification email,its account stays Unverified until an Admin approves it.
AC-02	An Admin can approve or reject a pending company,a rejection requires a reason and the company is notified.
AC-03	An unverified company can browse listings but cannot buy, bid, or create RFQs and offers.
AC-04	A Seller can create a listing with all required fields; the listing is not visible publicly until an Admin approves it.
AC-05	When a verified Buyer purchases a fixed-price listing, it is reserved immediately and no other buyer can purchase it.
AC-06	When the Buyer confirms receipt, the Transaction becomes Completed, and the simulated payment is processed according to the approved commission policy.
AC-07	Both parties can see the Transaction status at every stage.

AC-08	A Buyer cannot bid without a held deposit; a bid below the minimum or after the end time is rejected; a valid bid is acknowledged in under 1 second.
AC-09	Keyword search and filtering by category, condition, location, and price return correct results in under 2 seconds.
AC-10	At the end time the auction closes automatically, the highest valid bid is marked Won, and a Transaction is created for the winner.
AC-11	Deposits are released or refunded according to the auction outcome.
AC-12	A Buyer can create an RFQ, Sellers can submit offers before the deadline, and the Buyer can select exactly one offer; selecting it creates a Transaction and marks the other offers Not Selected.
AC-13	A cancelled or expired RFQ accepts no more offers.
AC-14	A dispute can be filed only on an existing Transaction; the Admin's resolution updates the Dispute and Transaction statuses.
AC-15	A suspended company cannot sign in or transact.
AC-16	Key events (verification decision, purchase, outbid, offer selected, dispute resolution) send an in-app notification and an email to the affected party.
AC-17	Passwords are stored hashed, and administration pages are not accessible to non-Admin users.
AC-18	Verified companies can complete all three transaction paths end to end (Fixed Price, Auction, RFQ) with simulated payments, Admin moderation, and transaction records.

Conclusion

AssetX attacks a real, evidenced gap — fragmented B2B asset trading — with a focused MVP that proves strangers can complete trusted deals online. The team validates first with ten companies and simulated money, then scales what the pilot proves.

> ***“Validate first. Scale second.”***

End of document.